

Coupled SWAT-MODFLOW model for assessing watershed responses to groundwater sustainability: A novel ensemble approach

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ABSTRACT

Study Region: Gilgel Gibe Watershed, Omo Gibe River Basin, Ethiopia.

Study Focus: Groundwater depletion adversely affects ecosystem health, leading to reduced vegetation cover, changes in habitat suitability, and increased vulnerability to environmental stress. Quantifying groundwater sustainability under surface water-groundwater interactions is crucial for understanding system responses to anthropogenic changes. Hence, the main objective of this study was to quantify groundwater sustainability using a novel ensemble approach of the coupled SWAT-MODFLOW model. The performance of the integrated SWAT-MODFLOW model was assessed using the Parameter Estimation Tool (PEST) with streamflow and aquifer properties, and was found to be in reasonable agreement between the predicted and observed values.

New hydrological insights: Annual ensemble values of exploitable amounts of groundwater discharge, minimum desirable flow, and groundwater recharge were used to compute the groundwater footprint and sustainability. The overall basin-wide groundwater footprint was moderately depleted, revealing the need for effective management strategies to mitigate long-term groundwater decline and vulnerability to groundwater droughts. This has resulted in food insecurity due to diminished crop yields and shifting agricultural practices. Overall, this study offers valuable insights for developing informed policy frameworks that prioritise resilient watershed management to support the achievement of sustainable development goals.

1. Introduction

Humans encounter stress and vulnerabilities in both groundwater and surface water resources (Haddeland et al., 2014). Rapid population growth and economic expansion, coupled with inadequate human-induced water management, are worsening freshwater scarcity and increasing demand (Abraha et al., 2022; Adane et al., 2021; Liu et al., 2017; McDonald et al., 2011). Groundwater, which constitutes approximately 99 % of the Earth's available liquid freshwater, has vast potential benefits in the social, economic, and environmental aspects of human consumption (United Nations, 2022). Groundwater is an invisible yet crucial resource that must be utilised and effectively managed to meet future demands (Aeschbach-Hertig and Gleeson, 2012; West et al., 2023). Unfortunately, it is

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often misunderstood, undervalued, and mismanaged (Braune and Xu, 2010; Foster et al., 2013; United Nations, 2022), and requires a paradigm shift to address sustainable management strategies (Elshall et al., 2020).

Both surface and groundwater resources are highly affected by uncontrolled human activities (Ashraf et al., 2017; Birhanu et al., 2021). In drylands, groundwater recharge occurs when precipitation accumulates in the surface water bodies (Yao et al., 2018). Nevertheless, drought conditions restrict aquifer recharge, leading to increased demand and a reduction in groundwater storage (Fleckenstein et al., 2006). Groundwater development can significantly impact streamflow because the water exchange between streams and aquifers is governed by variations in the hydraulic head (Winter, 2007). Resilient hydrological systems sustain healthy freshwater ecosystems, offer goods and services to society, and maintain biodiversity under natural and human disturbances (Alley et al., 1999; Mays, 2013; Richter et al., 2003; Thomas et al., 2019). However, decision-making processes often encounter difficulties owing to limited data availability and uncertainty (Healy, 2019; Wada, 2016). Groundwater and surface water, as integrated freshwater resources, are essential for the well-being of society and aquatic ecosystems (Vörösmarty et al., 2018; Winter et al., 1999) and enhance sustainable development frameworks (Bredehoeft, 2002). Sustainable resource management requires a thorough understanding of the effective use of storage, long-term yield, and protection of aquatic environments in eco-friendly settings (Gejl et al., 2019; Gorelick and Zheng, 2015; Mays, 2013). Hence, a novel and pivotal strategy is required to quantify and qualify surface water-groundwater interactions (SGWI) to address water stress.

The effects of SGWI at the regional scale require an ample understanding of aquifer production for long-term consumption. Integrated SGWI models enable the holistic characterisation of complex nonlinear relationships in watershed systems (Barthel and Banzhaf, 2016; Newcomb and Godsey, 2023). Despite the challenging nature of SGWI modelling, analysing spatiotemporal effects is necessary to evaluate flux exchanges, including hydrological extremes and management practices (Aliyari et al., 2021; Geris et al., 2022; Sith et al., 2019), which, in turn, facilitates the development of practical management policies for sustainable resource utilisation, which is crucial for developing new findings to support water management (Ahmadi, 2023; Rassam et al., 2013). Different SGWI models have undergone significant refinement and enhancement, addressing exchanges that are either fully or loosely coupled (Ahmadi, 2023). The integration of the Soil and Water Assessment Tool (SWAT) (Arnold et al., 1998) with the modular three-dimensional finite-difference groundwater flow model (MODFLOW) (McDonald and Harbaugh, 1988) has attracted significant attention. SWAT-MODFLOW demonstrates the exchange of flux between the land surface and unsaturated zone of the aquifer system, providing a complete understanding of feedback mechanisms (Aliyari et al., 2019; Bailey et al., 2016; Guzman et al., 2015; Kim et al., 2008; Sophocleous and Perkins, 2000), overcoming the constraints imposed by separate models to represent the SGWI processes.

Although different studies have reported the applicability of modelling SGWI at spatiotemporal scales in headwater regions globally (Guevara-Ochoa et al., 2024; Taie Semiromi and Koch, 2019; Wang and Chen, 2021; Yifru et al., 2022), most have failed to assess the potential effects on groundwater resource sustainability. However, in these regions, aquifers experience temporal fluctuations with diminished low flow due to groundwater extraction (Imberger et al., 2023; Winter, 2007), affecting the SGWI, which is crucial for defining hydrologic regime variability (Richardson and Danehy, 2007; Richter et al., 1998) and ecosystem functions (Piczak et al., 2023; Poff et al., 1997; Richter and Thomas, 2007). Anthropogenic impacts on ecosystem health shape the sustainable consumption of water resources (Hoekstra and Mekonnen, 2012; Pérez et al., 2019). Hence, integrating the water footprint with effective water governance is essential for monitoring human stress on freshwater consumption (D'Ambrosio et al., 2020), although the lack of reliable data challenges robust estimations (Gleeson and Wada, 2013; Gohari et al., 2013). Sustainable groundwater consumption was assessed to determine water stress using a global estimate based on the aquifer water balance (Gleeson et al., 2012). Although this concept reduces uncertainties in groundwater resource assessment and improves the formulation of management strategies (Mahdavi, 2021), global estimates are more prone to errors (Moeck et al., 2020) and neglect aquifer physical processes (Gleeson et al., 2020). Additionally, it can be estimated using data-driven models for an optimal water supply (Ghordoyee Milan et al., 2023; Kourgialas et al., 2018). However, this approach often ignores the physical processes occurring in the aquifer subsurface. This leads to an inadequate allocation of resources and fails to consider the impacts of groundwater depletion, resulting in unreliable water resource planning.

Recently, integrated hydrological modelling has become a cornerstone for understanding the dynamics of SGWI in Ethiopia, particularly in regions where aquifers provide a critical buffering capacity against seasonal water shortages (Mengistu et al., 2021). Despite these advances, recent studies have important methodological and conceptual limitations. (Tigabu et al., 2020) employed a SWAT-MODFLOW framework to quantify spatiotemporal dynamics in the Lake Tana Basin, demonstrating strong hydraulic connectivity between aquifers and river networks. While this study clarified how groundwater discharge sustains streamflow during dry seasons, it stopped short of sustainability thresholds, leaving decision-makers without clear guidance on where groundwater management interventions are most urgently needed. (Yifru et al., 2021) assessed the impacts of human activities and climate change in the Meki catchment, without exploring localized vulnerability, where reduced recharge could intersect with concentrated pumping to create unsustainable drawdown hotspots. Furthermore, this study did not evaluate whether recharge deficits would be reversible or lead to long-lasting depletion, which is essential for long-term aquifer management. In the same catchment, (Taye et al., 2022) evaluated irrigation sustainability and found that existing water supplies are insufficient to meet seasonal agricultural demands, although the SGWI could partially offset deficits. Although the study highlights the critical role of groundwater, it does not quantify how expanded groundwater use would alter the long-term storage balance or affect downstream flows and ecosystems. (Yifru et al., 2022) also integrated nitrate transport modelling to evaluate pollution under urban expansion in the Akaki catchment, an important step toward linking hydrologic modelling with water quality outcomes. However, the focus remained on pollutant transport without examining how nitrate contamination interacts with quantity depletion through reduced dilution capacity, overlooking the important cumulative effects on aquifer vulnerability and sustainability.

Across these studies, the absence of a groundwater footprint has been identified to determine where escalation would be most ecologically and hydrologically impactful. While each study offers valuable insights into specific water resource conditions, they have

the shortcoming of capturing the full complexity of the environmental interactions that shape groundwater sustainability in rapidly changing watersheds. Frameworks that address the combined effects of anthropogenic activities on SGWI must be developed. Understanding watershed processes using an integrated modelling framework is crucial for developing reliable and sustainable water management strategies. However, regional-scale SGWI responses to sustainable groundwater consumption remain unclear. No studies have focused on assessing the groundwater footprint (GF) while accounting for the synergistic effects of the SGWI. The development of new perspectives for groundwater footprint evaluation is necessary to improve the reliability of simulations and enable long-term decision-making. However, there is no evidence quantifying GF under regional-scale SGWI conditions using the coupled SWAT-MODFLOW model. Thus, the main purpose of the present study was to quantify groundwater sustainability responses to the GF using an ensemble approach based on the coupled SWAT-MODFLOW model. Estimating the GF scrutinises watershed responses to long-term groundwater utilisation and introduces a novel framework that is valuable for establishing a policy framework for a proper and sustainable water management plan. The present study in the Gilgel Gibe watershed builds on this foundation by applying SWAT-MODFLOW, emphasising ensemble analysis and integrated evaluation of groundwater sustainability, thereby addressing important research gaps and providing new insights into SGWI. We demonstrate how recharge and exchange fluxes contribute to the long-term sustainability of groundwater resources, offering novel hydrographic and groundwater management insights.

2. Materials and methods

2.1. Study area

Gilgel Gibe River Watershed (Fig. 1) is located in the Omo Gibe River Basin, Ethiopia. Most groundwater infiltrates directly from precipitation and flows laterally in regional drainage basins by rivers in valleys or vertically, recharging deeper aquifers in unconsolidated sediments and volcanic rocks. Lateral discharge contributes to aquifer formation in unconsolidated sediments. This region is predominantly marked by a flat elevated plateau, with summits exhibiting slopes ranging from mild to severe slopes. The Quaternary stratified alluvial deposits were a result of rifting, followed by downward warping and the creation of distinct grabens in the basin. Infiltration is most favourable in plateaus and is characterised by substantial, alluvial deposits. The lithology of the aquifer consists of sandy pyroclastic deposits from the Pleistocene-Holocene volcanic period, with lower felsic volcanic and sedimentary formations and higher felsic volcanic deposits. In addition, there is a highly permeable layer of volcanic-sand. Quaternary alluvial soil has promising prospects for deposits and structures characterised by enhanced permeability. Fractured Quaternary volcanic formations generate significant groundwater flow. Quaternary alluvial deposits are found in wide valleys on plateaus and highland regions. Nevertheless, the available quantitative data on watersheds are insufficient or irregularly scattered, hindering comprehensive quantitative studies. However, the quantitative data on the watershed are inadequate or evenly distributed enough to allow for a detailed quantitative analysis. For this reason (Mengistu et al., 2022a), productive aquifer zones were produced using proxy data as a pilot understanding of

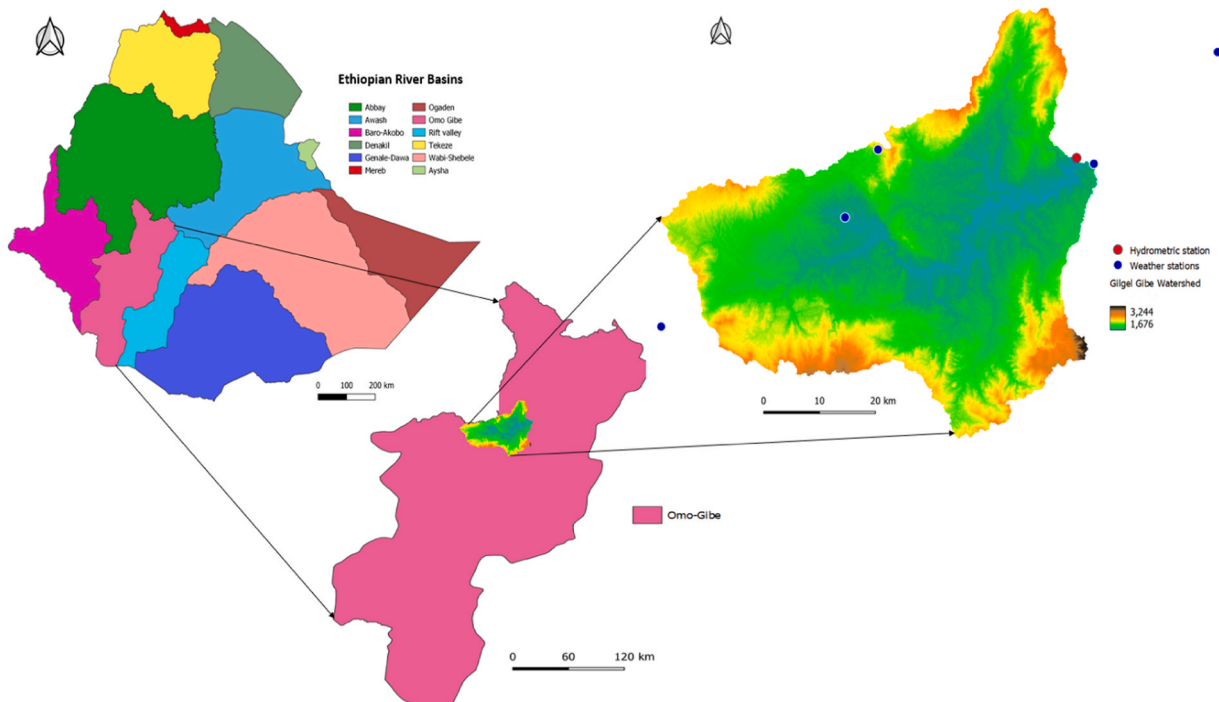


Fig. 1. Location Map of the study area.

the recharging conditions. Most river channels follow lineament and drainage density patterns. Lithology and land use factors control the groundwater flow process, and recharging effects increase deep soil water percolation.

2.2. Input Datasets

In hydrological modelling and water resource studies, assessing the reliability and quality of historical climate data records is required. The SWAT model requires elevation, land use/land cover (LULC), soil, and daily meteorological data to simulate watershed hydrologic processes. A Digital Elevation Model (DEM), a numerical depiction of topographical characteristics, was extracted from the USGS Shuttle Radar Topography Mission (SRTM) (<https://earthexplorer.usgs.gov>). Soil characteristics and land use information are essential for modelling hydrological components. Soil data were obtained from the FAO soil database. Land use/land cover data were extracted from high-resolution Landsat images from the GlobeLand30 collection (<http://globeland30.org>). SWAT requires daily weather data (precipitation, temperature, relative humidity, solar radiation, and wind speed). The required weather data were collected from the National Meteorological Agency of Ethiopia. However, the spatiotemporal variability of rainfall in the country is high, and there exist various non-homogeneities in climatic data series, reflecting the source of uncertainty, which results in poor model outcomes. In addition, the continuity of recorded data was missed due to the absence of an observer or/and failed instruments. The observed daily recorded weather data were checked for missing data and continuity before using any hydrologic modelling. The techniques used to fill in missing data depend on the percentage of missed data and the optimal location of the neighbouring stations. A multivariate statistical approach was used to fill the missing data and verify the overall data quality and reliability for model development and application. The streamflow data required for model calibration and validation were obtained from the Ethiopian Ministry of Water Resources, Irrigation, and Electricity (MoWIE) from the watershed outlet (Mengistu et al., 2022b).

2.3. Land Surface Model

The Soil and Water Assessment Tool (SWAT) model (Arnold et al., 1998) is a hydrological model that operates on a continuous daily timescale and is designed to simulate hydrological processes on the land surface. It is designed to mimic many processes related to surface water flow, soil temperature, in-stream and soil domain activities, bacterial and pond storage, pesticides, land management, sediment, and nutrient transfers. The SWAT model imposes spatial discretisation of land use, soil, and topography to accurately reflect watershed hydrological processes. Hydrological Response Units (HRU) are formed by unique combinations of land use, soil, and slope features, which contribute to spatial heterogeneity and enhance the precision of loading predictions. The computation of all soil water balance elements was based on an HRU to show analogous hydrologic characteristics of the HRU (Arnold et al., 1998). The threshold values of the numerous HRUs, as specified by (Her et al., 2015), were used to enhance the simulation efficiency while preserving the important characteristics of the watershed without compromising the accuracy of the biophysical inputs (Femeena et al., 2022). To

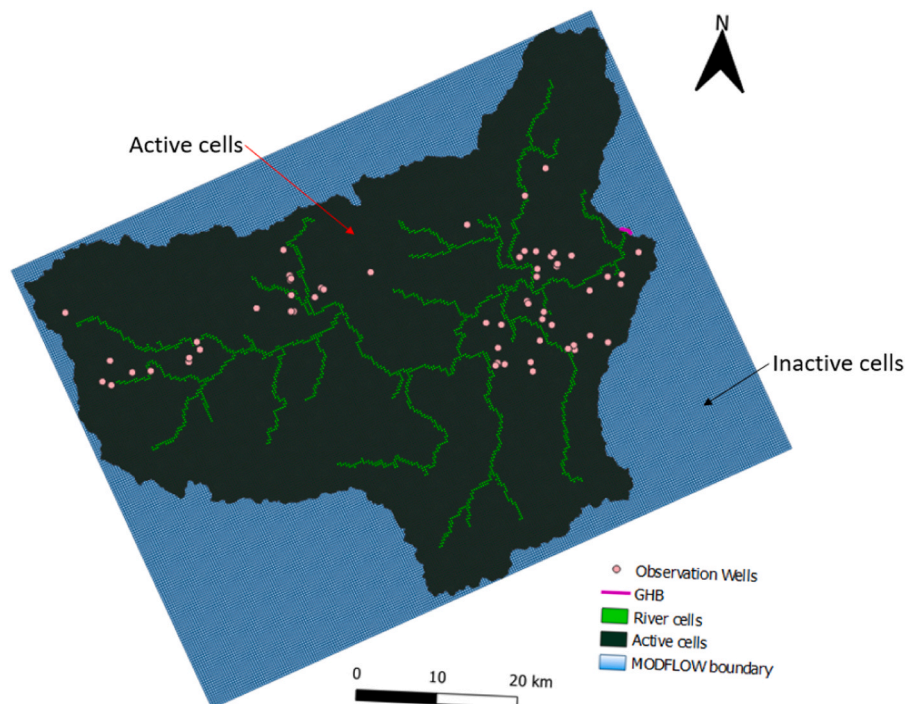


Fig. 2. Boundary conditions for GFM construction.

account for the water shortage and limited exchange across HRUs, the model used analytic delaying methods to incorporate the groundwater return flow from each HRU into the sub-watershed outputs (Mengistu et al., 2022b). The model replicates the dynamics of a shallow unconfined aquifer that supplies water to the mainstream or flows through the sub-basin, along with a confined deep aquifer, and is determined using the water balance equation (Eq.1).

$$SW_t = SW_o + \sum_{i=1}^t (R_{day} - Q_{SUR} - E_a - W_{seep} - Q_{GW}) \quad (1)$$

Where; SW_t final water content (mm), SW_o initial soil water content (mm), t time (days), R_{day} precipitation amount (mm), Q_{SUR} is the amount of surface runoff (mm), E_a actual evapotranspiration (mm), W_{seep} water entering to vadose zone from soil profile (mm), Q_{GW} return flow amount (mm).

2.4. Groundwater Flow Model

The numerical Groundwater Flow Model (GFM) helps interpret the available site information for long-term and efficient groundwater utilisation. In the present study, the GFM was developed to achieve hydrologic balance and identify the model domains imposed during SGWIs. A robust, physically based, distributed, Modular Three-Dimensional Finite-Difference Model (MODFLOW) (McDonald and Harbaugh, 1988) was used. MODFLOW solves groundwater flow problems of the groundwater head at the centre of the grid cells. In this study, the GFM was constructed using MODFLOW-NWT (Niswonger et al., 2011), which employs an advanced Newtonian solver formulation to address the computational challenges associated with the nonlinearities of drying and rewetting in unconfined groundwater-flow systems. This ensures the activation of all cells across the model domain with the Upstream Weighting (UPW) package (Hunt and Feinstein, 2012). MODFLOW-NWT is an appropriate tool for analysing the dynamics of SGWIs (Wöhling et al., 2018). To simulate various hydrologic processes, representative critical sources and sinks were defined to solve numerical GFM problems.

The groundwater aquifer and SWAT watershed were assumed to have the same boundary (approximately 2931 km²), that is, the surface of the three-dimensional computational grid, as shown in (Fig. 2). We ensured that the top surface of the model aligned with the actual ground level. First, the collected borehole data were used to establish the depth of the monitoring stations, which extended from the surface of the ground to the base of the aquifer. To simulate various hydrologic processes, representative critical sources/sinks were defined to solve numerical GFM problems. The general head boundary is allocated to reflect water flowing into or out of the model cell. The surface water network for the model domain consisted of stream features represented by the river package obtained from the SWAT model (Mengistu et al., 2022b). The river stage was determined based on the model top of MODFLOW derived from the SWAT DEM.

We used a 0.05 head (L) and 0.006 flux (L³/T) tolerance during the NWT solver application. The specified option of model complexity corrects the groundwater head relative to the cell bottom altitude when the cell is surrounded by dry cells. Coefficients of 0.01 and 0.1 were used to maximise and minimise the weighted applied head change, respectively, with a momentum coefficient of 0.01. The residual control of 50 backtracks in the head changed between nonlinear iterations. In addition, a 0.9 proportional decrease in the root mean square error of the GW flow equation was used to determine whether the residual control was required at the end of a nonlinear iteration with a residual factor of 0.001 that reduced the head change between nonlinear iterations. The asymmetric solver options, the stabilised conjugate gradient solver variables were applied with reduced system preconditioning of 0.001 residual reduction convergence criteria with a 0.1-drop tolerance for preconditioning and head closure criteria for inner iterations. The river package simulates the water exchange between the aquifer and river through the riverbed with river stage, river bottom level, and streambed conductance.

Riverbeds enable accurate hydrologic propagation of stress indicators in aquifer systems. Base flow to a stream occurs from aquifers to streams following Darcy's law whenever the simulated water-table elevation is higher than the prescribed stream stage in a cell. Riverbed conductance, interpolated to the river grid cells, is crucial for calculating stream-aquifer interactions, as its spatial variability influences fluxes between streams and aquifers (Abimbola et al., 2020). The general head boundary was allocated to reflect the water flow into or out of the model cell. The surface water network for the model domain consisted of stream features represented by a river package. This package simulates river-aquifer exchange through the riverbed using the river stage, river bottom level, and streambed conductance. Riverbeds enable the accurate hydrologic propagation of stress indicators in aquifer systems. Riverbed conductance calculates stream-aquifer interactions because its spatial variability influences the fluxes between streams and aquifers (Abimbola et al., 2020). The well package simulates the operations of wells that either extract water from an aquifer or inject water into the aquifer. The volume of groundwater removed by evapotranspiration was calculated as a linearly varying function of the depth below the water table using an evapotranspiration package. The recharge package simulates the recharge from land infiltration/inflow into the model domain. An unconfined aquifer and single-layer GFM were used to discretise the model domain, which had dimensions of 201 rows and 263 columns. The boundary conditions with the MODFLOW grid were implemented during the model construction.

The initial hydraulic conductivity, river conductance, and general head boundary conductance values were adjusted in the successive model run until the simulated heads matched the observed ones with a reasonable error of the calibration targets. The MODFLOW parameters of hydraulic conductivity (0.25 – 100 m/d), riverbed conductance (0.25 – 500 m/d), and static groundwater head (1.2 – 56 m) from observation wells were used. Because there are no recorded continuous groundwater level data, transient-state MODFLOW simulation was initiated after the simulated and observed weighted groundwater levels were in agreement prior to the coupling process. The riverbed conductivity interpolated to the river grid cells is vital for calculating stream-aquifer interactions. In an unconfined aquifer, MODFLOW efficiently predicts groundwater flow movements in the aquifer system based on mathematical

principles (Eq. 2) of mass conservation and Darcy's law (McDonald and Harbaugh, 1988).

$$\frac{\partial}{\partial x} \left(K_x \frac{\partial h}{\partial x} \right) + \frac{\partial}{\partial y} \left(K_y \frac{\partial h}{\partial y} \right) - W = (Ss) \frac{\partial h}{\partial t} \quad (2)$$

where the hydraulic head (h) is defined in the (x , y) Cartesian coordinate system, the horizontal and vertical hydraulic conductivities (K_x , K_y), (W) is the water exchange from the sources, and (Ss) is the specified dimensionless water storage.

2.5. Coupled SWAT-MODFLOW Model

A thorough hydrological system simulation requires an examination of the water balance components. The SWAT groundwater module cannot represent distributed aquifer parameters (Kim et al., 2008) and overlooks SGWIs, thereby limiting the representation of dynamic groundwater tables (Aliyari et al., 2019). The model has a simplified groundwater concept, resulting in low performance in base flow simulation. Owing to its quasi-distributed nature, SWAT inadequately represents aquifer heterogeneity and is unable to simulate groundwater discharge to streams or aquifer seepage (Pfannerstill et al., 2014). However, MODFLOW has limitations in terms of surface water flow evaluations and recharging predictions. The SWAT-MODFLOW model (Bailey et al., 2016) combines the hydrologic calculations of the MODFLOW-NWT (Niswonger et al., 2011) and SWAT (Arnold et al., 1998) models into an integrated package, allowing for the advancement of their respective applications.

2.5.1. Geospatial Referencing of SWAT-MODFLOW

When combining the SWAT and MODFLOW models, the variables between the river formations and the SWAT stream network location are predictable for the MODFLOW cells (Guzman et al., 2015) for the geospatial reference of the SGWI. MODFLOW quantifies the amount of groundwater released into streams, whereas SWAT quantifies the amount of water that flows over the land surface and through the soil into the streams. Although the SWAT groundwater module cannot provide the geographical distribution of the exchange features of the HRUs using MODFLOW cells, this can be achieved by utilising the combined SWAT-MODFLOW system. Using a gridded spatial discretisation of groundwater discharge and considering aquifer characteristics, SWAT-MODFLOW models the three-dimensional groundwater discharge dynamics at the saturation level. The linkage between SWAT and MODFLOW was established by recharging SWAT-MODFLOW at grid cells using percolation estimates that were computed for each SWAT HRU. Using the MODFLOW recharge package, the percolation fluxes modelled at the HRU level in the SWAT model were transferred to the aquifer domain. To define cell relationships with MODFLOW grid cells, HRUs were spatially disaggregated into Disaggregated HRUs (DHRUs). To connect SWAT and MODFLOW, DHRUs geolocated certain polygons. Data were sent via deep percolation from the DHRUs to the associated MODFLOW grid cells. MODFLOW transfers groundwater discharge from each sub-basin to SWAT using deep aquifer recharge provided by SWAT to each grid cell. Streambed flux exchange is a key component of MODFLOW aquifer and river cell water-flow computations. To ensure that the right sub-basin channels received the MODFLOW-computed SGWI rates for streamflow channelling, the stream cells were categorised based on the SWAT sub-basin (Bailey et al., 2016).

2.5.2. Water Resources Fluxes Under River-Aquifer Interaction

Quantifying river-aquifer interaction fluxes is essential for effective water resource management. The transfer of water between water bodies and underlying aquifers includes the movement of recharge and pollutants (Bailey et al., 2016). The SGWI was calculated as the difference between the water level and hydraulic conductivity of the riverbed. The length of the MODFLOW River corresponds to the length of the main channel in SWAT. The MODFLOW river stage for each river cell was calculated from the SWAT channel depth to determine the SGWI in each SWAT subbasin. In watershed management, simulating water resource fluxes under stream-aquifer interactions using an integrated model provides more accurate information than separate models. Hence, an integrated SWAT-MODFLOW model was employed to evaluate the influence of the hydrological responses of watersheds on the groundwater footprint under regional SGWI. The coupled model uses the MODFLOW river package to compute water exchanges through the aquifer and streams using (Eq. 3).

$$Q_{\text{flux}} = (L)(K)(W) \left(\frac{h_{\text{riv}} - h_{\text{gw}}}{T_{\text{riv}}} \right) \quad (3)$$

Where Q is Darcy's volumetric flux [L^3/T], H_{riv} is the river stage [L], h_{gw} is the hydraulic head of the adjacent groundwater [L], K is the hydraulic conductivity of the riverbed material [L/T], L is the length of the reach [L], W is the width of the river channel [L], and T_{riv} is the thickness of the riverbed [L].

2.6. Model Performance Analysis

2.6.1. Calibration of GFM

GFM calibration involves determining a set of BCs and hydrogeological parameters that produce measured and simulated hydraulic heads within a known range of errors (Anderson and Woessner, 1994). The GFM was developed as a single-layer aquifer because of the lack of accurate aquifer stratigraphy data. This could result in unreliable estimations of water table profiles, introducing uncertainty in aquifer properties. Despite the challenges posed by the absence of aquifer parameters and sufficient observational datasets, GFM calibration provides valuable insights for mitigating model uncertainties. The calibration quality depends on the amount and

reliability of the available field data. However, using realistic parameters to adjust the calibration process reduces the problem of nonuniqueness. Based on the impact of static water level variations, we calibrated a steady-state MODFLOW simulation using the average groundwater level from the observation wells. Nevertheless, the stress periods were attenuated in the simulation of the steady state model.

Owing to the scarcity of long-term groundwater level data, the GFM was not calibrated for transient conditions. GFM errors may arise from the assumptions made and aquifer conditions. In this study, the model was calibrated using the auto-calibration method of the Parameter Estimation Tool (PEST) (Doherty and Johnston, 2003) to reproduce the observed static heads. In model evaluation, auto-calibration minimises the uncertainties associated with subjectivity. The optimal model value was achieved using PEST and trial-and-error to reduce the total error transfer on a single parameter under uneven calibration conditions. Aquifer parameter adjustments were made to minimise the discrepancy between the predicted and observed water levels. In addition, a trial-and-error approach was used to incorporate subjective judgment and understand the model's behaviour. The coefficient of determination (R^2) was evaluated using a scatter plot of the measured and simulated heads. The head residuals in the GFM refer to the differences between the calculated and observed hydraulic head values at specific locations within the aquifer. The hydraulic head represents the groundwater potential energy, including the elevation and pressure components. In the GFM calibration, error assessment metrics were used to evaluate the accuracy of the GFM predictions relative to the observed values. ME represents the average difference between the model prediction and observation, with a negative ME indicating a consistent underestimation. The Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Error (ME) were used to determine the errors of the observed and simulated heads and were calculated as follows (Eqs. 4 – 6):

$$ME = \frac{1}{n} \sum_{i=0}^n (h_o - h_s) \quad (4)$$

$$MAE = \frac{1}{n} \sum_{i=0}^n |(h_o - h_s)| \quad (5)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (h_s - h_o)^2} \quad (6)$$

where h_o and h_s are the observed and simulated heads, respectively, and n is the number of observation wells.

2.6.2. SWAT and SWAT-MODFLOW Calibration

Calibrating the hydrological model enhances the accuracy of the simulations by reducing the uncertainty caused by errors in the interpretation of the input data. The SWAT model is a robust and interdisciplinary watershed model that is crucial for streamflow predictions. Identifying suitable parameters with optimal goal functions is essential for model performance (Moriasi et al., 2015). In data-scarce environments, parameters derived from gauged sites may need to be adjusted for ungauged sites during calibration owing to human actions (Mengistu et al., 2019). Accurate parameterisation expedites model calibration, enhances accuracy, reduces uncertainty, and elucidates hydrological processes. The SWAT model Calibration and Uncertainty Procedures (SWAT-CUP) with the Sequential Uncertainty Fitting Version 2 algorithm (SUFI-2) (Abbaspour et al., 2007) were used to calibrate and validate the model. SUFI-2 comprehensively addresses water resource uncertainty drivers, and a sensitivity analysis evaluates the model response to input

Table 1
Parameters used during the calibration of the coupled SWAT-MODFLOW model.

Parameters	Ranges
Runoff curve number (CN2)	−0.2 – 0.2
Base flow recession coefficient (ALPHA_BF, days)	0 – 1
Threshold depth of water in the shallow aquifer required for return flow (GWQMN, mm)	0 – 2000
Groundwater revap coefficient (GW_REVAP, mm)	0.02 – 0.2
Deep aquifer percolation fraction (RCHRG_DP)	0 – 1
Threshold depth of water in the shallow aquifer (REVAPMN, mm)	100 – 1500
Manning's N value for overland flow (OV_N)	0 – 1
Surface runoff lag coefficient (SURLAG, days)	1 – 20
Maximum canopy storage (CANMX, mm)	0 – 100
Groundwater delays (GW_DELAY, days)	10 – 500
Soil evaporation compensation factor (ESCO)	0 – 1
Effective hydraulic conductivity in main channel alluvium (CH_K2, mm/hr)	0 – 100
Plant uptake compensation factor (EPCO)	0 – 1
Available water capacity of the soil layer (SOL_AWC, mm H2O/mm soil)	−0.5 – 0.5
Hydraulic conductivity of soil (SOL_K, mm/hr)	−0.5 – 0.5
Aquifer hydraulic conductivity (K, m/day)	0.125 – 100
River conductance (COND, m ² /day)	0.25 – 100
Specific yield (Sy, %)	0.1 – 0.5
Specific storage (Ss, m ^{−1})	1×10^{-4} – 1×10^{-8}

adjustments by leveraging the parameter refinement (Abbaspour et al., 2015). The prediction accuracy of the SWAT model in the watershed was determined during separate calibration (1997–2006) and validation (2007–2014) periods.

Integrated model parameter optimisation processes were performed using PEST principles (Doherty and Johnston, 2003), assigning suitable ranges of aquifer parameters such as specific yield, specific storage, and hydraulic conductivity. This enables the simultaneous adjustment of parameters to ensure a realistic replication of coupled hydrological patterns. The negligible value of specific storage was given as the aquifer was unconfined, and the MODFLOW-NWT simulation resulted in a reasonable specific yield. The calibration used the initial parameter ranges (Table 1) for the gauged streamflow to establish representative parameters. Then, the performance indices of the Coefficient of Determination (R^2), Percent Bias (PBIAS), and Nash–Sutcliffe Efficiency (NSE) (Moriassi et al., 2007) were evaluated and calculated as follows (Eqs. 7–9).

$$R^2 = \frac{(\sum_{i=1}^n (Q_s - \bar{Q}_s)(Q_o - \bar{Q}_o)^2)}{(\sum_{i=1}^n (Q_s - \bar{Q}_s)^2 \sum_{i=1}^n (Q_o - \bar{Q}_o)^2)} \quad (7)$$

$$NSE = 1 - \left[\frac{\sum_{i=1}^n (Q_o - Q_s)^2}{\sum_{i=1}^n (Q_o - \bar{Q}_o)^2} \right] \quad (8)$$

$$PBIAS = \left[\frac{\sum_{i=1}^n (Q_o) - \sum_{i=1}^n (Q_s)}{\sum_{i=1}^n (Q_o)} \right] \times 100 \quad (9)$$

Where: Q_s (simulated value), Q_o (measured value), $\bar{Q}_s$ and $\bar{Q}_o$ (mean of simulated and observed streamflow, respectively).

2.7. Estimation of Groundwater Footprint

“Groundwater sustainability is defined as maintaining long-term, dynamically stable storage and flows of high-quality groundwater using inclusive, equitable, and long-term governance and management” (Gleeson et al., 2020). It is assessed through the estimation of the groundwater footprint (GF) and groundwater footprint index (GFI), which evaluate groundwater withdrawals relative to natural recharge and base flow. Estimating the GF based on regional-scale SGWI is indispensable for sustainable groundwater use and maintaining healthy groundwater-dependent ecosystem. The rate of aquifer depletion relative to its replenishment rate is referred to as groundwater stress. The sustainable use of groundwater is defined as the maximum amount of groundwater that can be drawn from an aquifer without causing depletion, which is determined by estimating groundwater recharge. Aquifers can be used to their maximum potential for sustainable production if the amount of water removed does not significantly deplete the local ecology and the minimum necessary flow in adjacent waterways is preserved (Sophocleous, 2000; Zhou, 2009). Sustaining groundwater resources relies on understanding the characteristics and magnitude of the GF, which compels them to remain below the maximum-acceptable level. The GF is required to evaluate groundwater usage and groundwater-dependent ecosystem services over an area, providing management insights (Gleeson et al., 2012).

A novel ensemble approach was applied to estimate the GF and stress within a coupled SWAT-MODFLOW framework, capturing these sustainability indicators by simulating SGWI, recharge processes, and baseflow contributions across the aquifer to support sustainable groundwater management. In GF computation, significant uncertainty stems from various sources, which diminishes with reduced observation uncertainty, ultimately affecting the accuracy of groundwater recharge estimates (Xie et al., 2018). Our study improves groundwater recharge estimation and introduces a novel approach for assessing GF accounting SGWI in determining groundwater stress. The model provides estimates at both the HRU level and MODFLOW grid cells, and the total annual average groundwater recharge over the watershed. However, groundwater extraction directly affects the hydrological regime of the associated aquifer, minimising recharge and reducing base flow, and should not exceed recharge to supply aquatic environments (Akbar et al., 2022).

Exploitable groundwater is the annual amount of water abstracted from an aquifer, accounting for socioeconomic and environmental constraints (Sophocleous, 2000). The area-averaged annual groundwater abstraction can be directly derived from the model area based on the specific yield and expected water level decline in dry periods to sustain the ecosystem. However, the exploitable groundwater resources were obtained as the rate of annual average groundwater discharge, excluding the minimum desirable flow that can sustain the environment during low-flow periods. To address the inherent uncertainties in the model parameters, input data, and system representations, ensemble estimation is utilised, combining model outputs to enhance the accuracy of the ensemble GF estimation, minimise individual model biases, and improve overall robustness. The ensemble integration method averages the predictions from the optimised model configurations, each representing different assumptions. This process produces a more reliable and comprehensive estimate, providing a more accurate assessment of groundwater stress. This approach improves the reliability of GF predictions, offering valuable insights for informed groundwater management and policy development regarding SGWI processes. GF and stress were determined using Eqs. (10–11).

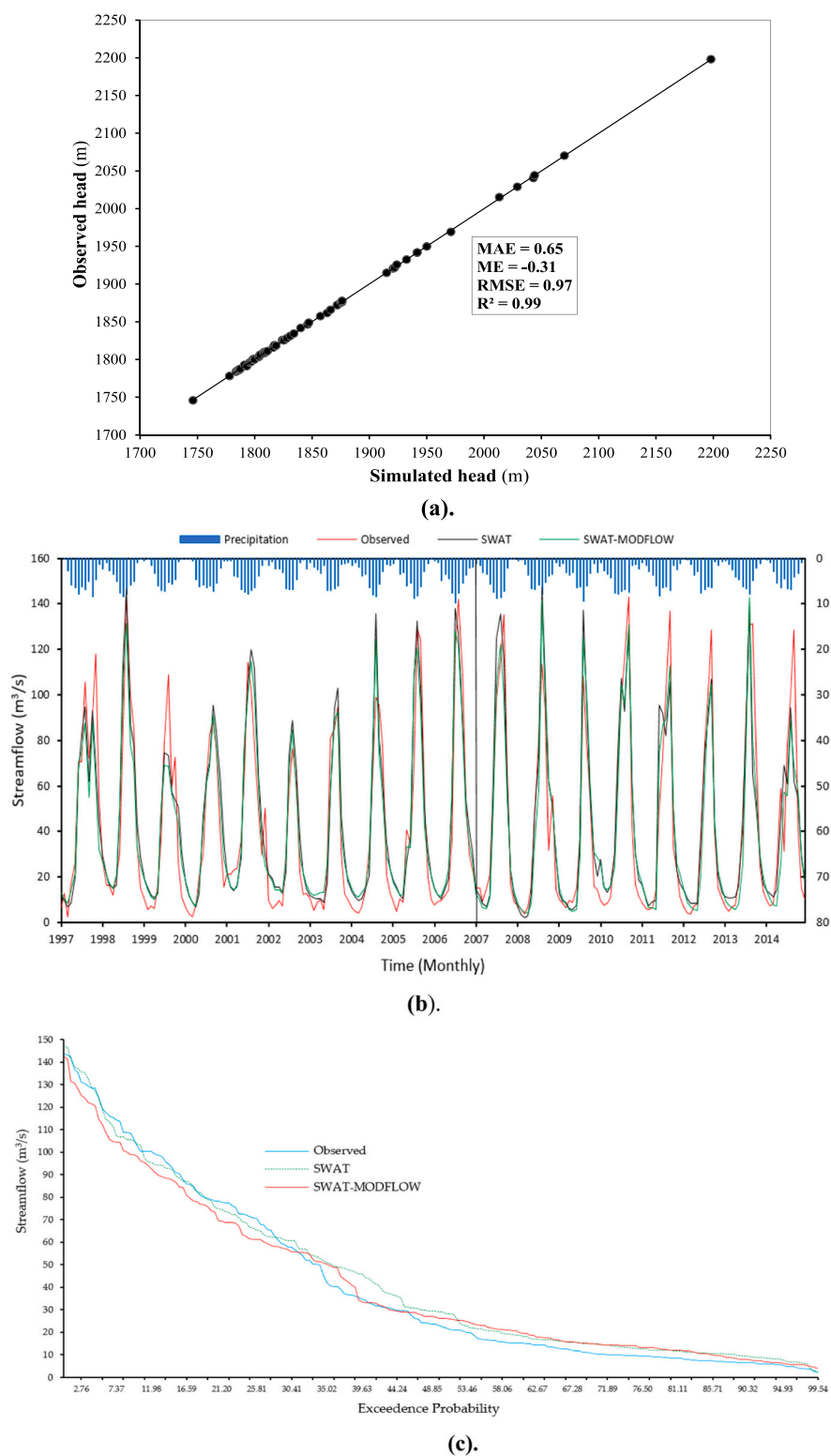


Fig. 3. (a). Simulated and observed static groundwater head (b). Hydrograph for SWAT and coupled SWAT-MODFLOW model simulations (C). Observed and simulated flow duration curves.

$$GF = \left[A_a \frac{C_a}{R_a - BF_a} \right] \quad (10)$$

$$GFI = \left[\frac{GF}{A_a} \right] \quad (11)$$

Where GF is the groundwater footprint, C_a is the average annual exploitable groundwater from the aquifer, R_a is the average annual groundwater recharge, BF_a is the average annual baseflow over the aquifer area, and GFI is the groundwater footprint index (groundwater stress over the A_a -is aquifer area (km^2)).

Aquifers with high GF undergo unsustainable withdrawals and depletion, as evidenced by the groundwater stress. The groundwater footprint is an approach that determines the equilibrium level of water in aquifers. A groundwater stress ratio greater than 1 indicates that the amount of water removed is larger than the amount put in, which indicates a state of stress on the aquifer. A ratio significantly greater than 1 suggests unsustainable withdrawals, often from past climatic conditions, surface water, and ecological systems dependent on groundwater. A GFI below 1 indicates sustainable extraction without groundwater depletion, meaning that groundwater use does not exceed the recharge capacity. Insights from groundwater footprints and stress indicators can inform policy design and water management strategies and evaluate potential mitigation strategies.

3. Results and discussion

3.1. Model calibration

Hydrological models must be accurately evaluated to reproduce affordable outputs. Assessing the SGWI leads to a calibration that captures groundwater processes, thereby improving low flow predictions (Rassam et al., 2013). A thorough calibration method ensures that the model parameters are adjusted to accurately represent hydrological system dynamics. Using a scatter plot and a correlation coefficient close to 1, the simulated and observed heads were examined using the MODFLOW simulation (Fig. 3a). Across the model domain, the head residuals indicated better agreement in representing the subsurface flow dynamics. The error assessments of ME, MAE, and RMSE were -0.31 , 0.67 , and 0.97 , respectively, indicating an acceptable fit for the data. An ME of -0.31 illustrated a systematic bias towards underestimation. MAE measures the average magnitude of errors, regardless of direction, and an MAE of 0.67 indicates an absolute difference between the model predictions and observations. The RMSE provides a measure of the typical deviation of the model predictions from the observed values. An RMSE of 0.97 implies that, on average, the GFM predictions deviated from the observations by approximately 0.97 units. The negative ME suggests a steady bias in the GFM predictions, whereas the MAE and RMSE provide supplementary measures of accuracy by considering both the magnitude and direction of the errors. Furthermore, the maximum absolute values of the model residuals were less than 10 % of the total head variation. In addition, the SWAT model demonstrates high accuracy, efficiency, and minimum bias in accurately representing complex hydrological processes (Mengistu et al., 2022b). Hence, the SWAT model is a helpful tool for surface water management, aiding sustainable decision-making that effectively reflects the intricate interactions between soil, water, and land use in the watershed. Hence, we anticipated that the GFM would be ready for coupling with the SWAT model to assess groundwater sustainability under stream-aquifer interactions.

The integrated SWAT-MODFLOW simulation performance was calibrated using the required streamflow and groundwater heads. However, only streamflow data were used because groundwater level time-series data were not available. The coupled model was calibrated using PEST, in which both SWAT and MODFLOW parameters were adjusted simultaneously. The PEST framework is a robust optimisation technique that iteratively adjusts the model parameters to reduce discrepancies between the observed and simulated variables. In the calibration process of the SWAT-MODFLOW model, various statistical indices, such as SWAT support, are significant indicators of the model performance in assessing watershed hydrologic responses. The statistical index results showed that the NSE was 87 %, R^2 was 84 %, and PBIAS was -8.73 . The NSE value of 87 % indicates that the model effectively replicated 87 % of the variability observed in the data, demonstrating a strong agreement between the model predictions and observed hydrological responses. Similarly, 84 % of R^2 indicates that the model explains 84 % of the variance in the observed data, further confirming the high degree of agreement between the model predictions and observations. Moreover, the PBIAS of -8.73 % revealed a slight tendency to underestimate the observed data, indicating a minor negative bias in the predictions. Overall, these statistical indices revealed the robustness of the integrated SWAT-MODFLOW model in simulating hydrological responses related to estimating groundwater sustainability, providing valuable insights for effective water resource management and decision-making processes. The high NSE and R^2 values validated the effectiveness of the model in explaining the patterns of hydrological responses. Conversely, a negative PBIAS suggests a tendency to underestimate, indicating potential areas for refinement and improvement.

The simulated and observed streamflow hydrographs plotted as shown (Fig. 3b) demonstrate that SWAT-MODFLOW performed well, providing realistic outputs for assessing watershed responses under regional-scale SGWI. To effectively regulate streamflow during the dry season, it is critical to understand the base flow, which creates the majority of streamflow during this period. Nevertheless, owing to constraints in monitoring observed data, modelling methods have become predominant in examining subsurface flow (Santhi et al., 2008). Human activities significantly control base flow conditions, which affect seasonal variations in river flow. We used digital BFlow filtering techniques (Arnold and Allen, 1999) to assess the proportion of base flow in the stream flow. SWAT-MODFLOW improved the streamflow simulation performance relative to SWAT, specifically in low flow simulations (Kim et al., 2008). The SWAT and SWAT-MODFLOW models yielded better agreement with the measured values and accurately simulated the base flow.

In the streamflow hydrograph, runoff and base flow frequency were separated in various passes of the SWAT and SWAT-MODFLOW simulations. Different stages of hydrographs are considered within frameworks with adequate performance indicators to conduct accurate model evaluations. The SWAT–MODFLOW model and SWAT had percentage differences of (8.38, 10.67, and 13.33), and (4.88, 6.80, and 6.15) observed for the first, second, and third passes of base flow, respectively. The SWAT–MODFLOW model performed better than the SWAT model. This indicates that groundwater systems are considered in streamflow simulations in areas with active groundwater dynamics. We evaluated the influence of base flow factors on the hydrograph by incorporating an alteration linked to groundwater flow rather than levels. This supports the assessment of the accuracy of the models in representing surface water, complexifying the emphasis on movement towards wells and model boundaries. The model effectively replicated a range of flow characteristics, capturing both low and high flows while accounting for other important phases of the hydrograph.

Streamflow was examined by developing a flow-duration curve, which is useful for evaluating statistical performance metrics, as validated by (Pfannerstill et al., 2014). To determine the accuracy of the model, the RMSE was computed in two parts: one for the high-flow periods of the real hydrograph (Fig. 3c) and the other for the low-flow periods when groundwater was more significant. Therefore, a low RMSE indicates flows that fall between 70 % and 100 %. The mid-flow RMSE, on the other hand, is defined as flows that have a likelihood of flow exceedance that falls between 20 % and 70 %. The SGWI results consistently showed lower RMSE for both low and mid flows, which is a probable explanation for the higher simulation in SWAT-MODFLOW than in SWAT, indicating improved simulations. The developed flow duration curve showed that the coupled model plays a higher predicts stream discharge

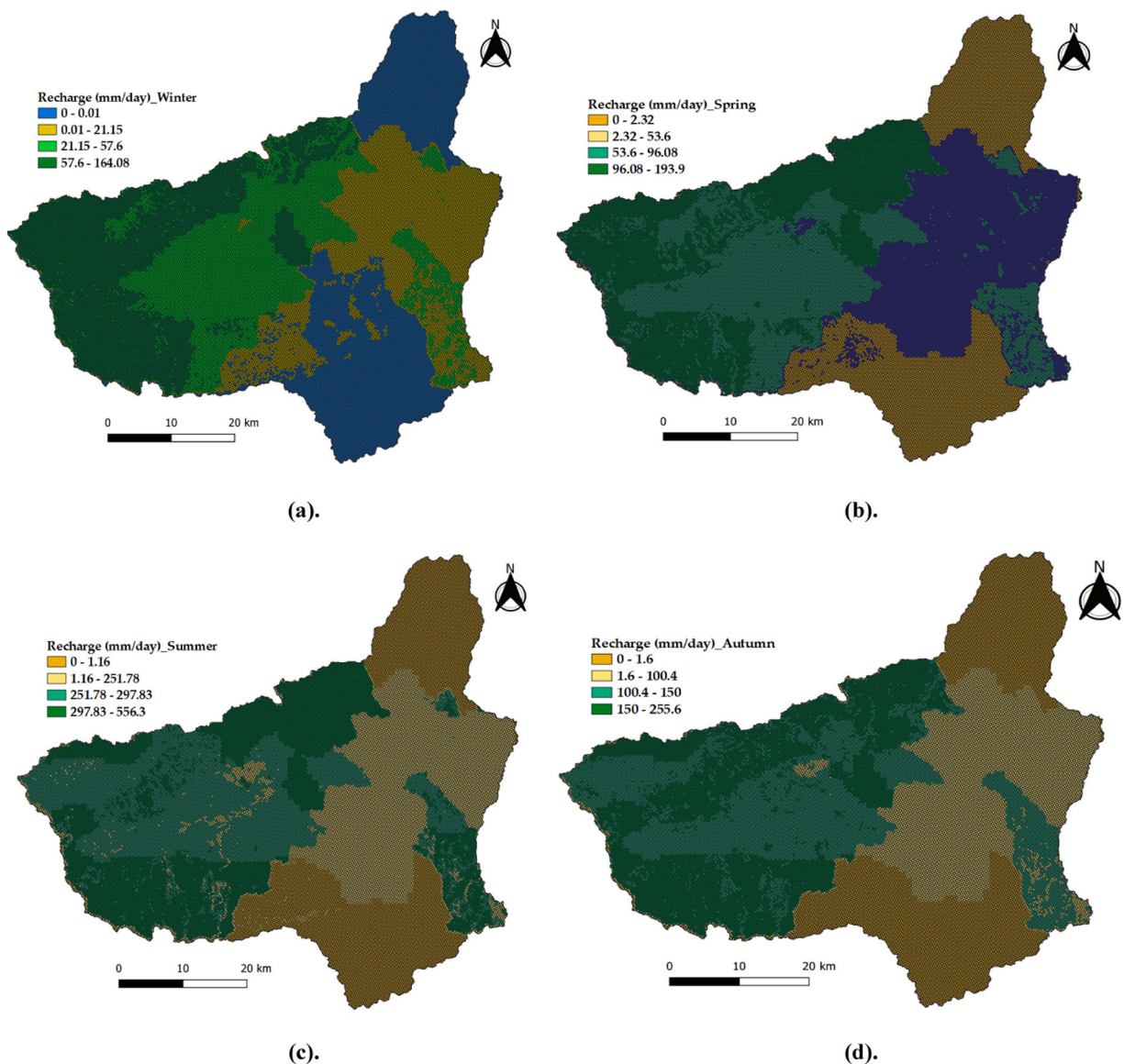


Fig. 4. Average seasonal spatiotemporal distribution of groundwater recharge.

better than the SWAT model. The simulation of SGWI enables a more accurate depiction of hydrological processes, resulting in more precise estimates of water availability and flux.

3.2. Estimating spatiotemporal groundwater recharge

Groundwater recharge is an essential hydrologic cycle responsible for maintaining aquifers that are replenished naturally by precipitation and artificially by human interaction. Groundwater recharge originates from various natural and anthropogenic events, such as soil infiltration capacity, climatic elements, and basin slopes. Natural-precipitation-induced groundwater recharge occurs on the exposed surfaces of aquifers. The simulated yearly water balances showed that evapotranspiration accounted for 44 % of the annual precipitation lost, whereas rainwater accounted for 56 % of the total rainfall that contributed to streamflow. The annual contribution of groundwater to precipitation was approximately 22.7 % on average. Groundwater and surface runoff accounted for 7.2 % and 3.32 % of the water output, respectively. Owing to the impact of topography, lateral flow accounts for 5.5 % of the average annual precipitation. The changes in groundwater recharge and river flow were proportional to rainfall and indicated that 25.22 % of the precipitation was lost as deep percolation. The maximum basin water yield occurred from (June to August) and the minimum from (December to March). (Rassam et al., 2013) scrutinised groundwater recharge seasonality, revealing significant global variations and emphasising its crucial role in water management and resource planning. Spatiotemporal variations in groundwater recharge significantly influence the hydrology of watersheds. These variations can trigger and exacerbate droughts or floods, directly affecting groundwater recharge, which is critical for groundwater storage.

Substantial spatiotemporal patterns of seasonal groundwater recharge variations were obtained using SWAT-MODFLOW, as shown in winter (Fig. 4a), spring (Fig. 4b), summer (Fig. 4c), and fall (Fig. 4d). Approximately 86 % of the groundwater recharge occurred in summer (June–September) and 14 % from (October–May). The maximum spatial rate of groundwater recharge occurs in the rainy season, and the minimum occurs during the dry season, which can profoundly influence water management. The results show that during (December – March), rainfall and evapotranspiration are prone to water deficits, and (June–August) have higher surface runoff rates which leads to flooding. Annual groundwater discharges are high (June–September) and low (December–March) owing to variations in evapotranspiration. Aquifer–river interactions involve either groundwater discharge or surface water inflow from a river into the groundwater system, depending on the disparity between the surrounding underground and surface water systems. The peak monthly percolation in July and August coincides with the maximum precipitation period. The lower exchange rate between groundwater and surface water in the lowlands may not have increased significantly from rainfall to enhance interactions with surface water. The highest magnitude of groundwater discharge occurred during summer (July–August). The magnitude of the simulated discharge varied for each river cell affected by the riverbed conductance. Wet seasons are crucial for sustaining stream yields during dry periods to balance water supply and environmental needs, as prolonged droughts reduce aquifer recharge. The prediction assessment of SGWI processes in wet (June – September) and dry (December – April) periods suggests that water resource management in an integrated and sustainable setting is necessary. Reductions in streamflow affect water temperature, and nutrient concentrations disturb aquatic life. Long-term decreases in streamflow affect flora along streams, which are valuable for preserving natural habitats and improving the water quality.

3.3. Groundwater discharge under stream-aquifer interactions

Groundwater discharge to streams and seepage from streams to aquifers along specific river reaches with significant SGWI were computed using the coupled model. Gaining conditions in the river network are driven by high groundwater levels that discharge directly into streams, whereas stream leakage reflects two-way exchanges between the rivers and aquifers. Discharge was highest during wet periods when groundwater levels were elevated and lowest during dry seasons. In the lowlands, the exchange rates remained low and did not increase significantly with the rainfall. The magnitude of groundwater discharge varied across the watershed, peaking between June and August and reaching a minimum in February. Discharge rates differed among river cells depending on riverbed conductance, and the overall aquifer recharge rates were lower than the discharge rates to the streams. Flat areas with shallow groundwater heads contributed the most to the stream gain. Some stream cells lost water to the aquifer due to withdrawals and elevation differences between the riverbed and aquifer interfaces, confirming significant SGWI.

Reduced freshwater discharge during droughts, combined with frequent pumping, lowers groundwater head, reduces streamflow, and limits groundwater storage replenishment. These declines threaten riparian vegetation, aquatic habitats, and water quality by altering flow regimes, temperature, and nutrient concentrations. Long-term streamflow reduction compromises the health and resilience of habitats. This study identified clear wet (June–September) and dry (December–April) periods, with wet seasons playing a critical role in sustaining streamflow during drought conditions in the dry season. Overall, the watershed was dominated by groundwater discharge to streams rather than recharge from rivers, underscoring the importance of comprehensive assessments that consider the SGWI processes. The spatial distribution of stream seepage has important implications for low-flow management (Wöhling et al., 2018). Therefore, integrated and sustainable water resource management is essential in this region.

3.4. Groundwater footprint and sustainability

The assessment of aquifer water balance and sustainable consumption based on the GF concept is critical for effective water resource planning. Groundwater recharge during the wet season ensures availability during droughts, thereby supporting a sustainable groundwater supply. Our study enhances recharge estimation and introduces a novel approach for assessing groundwater footprints by

accounting for SGWI using SWAT-MODFLOW to determine groundwater stress. The spatiotemporal groundwater recharge rate under the SGWI processes was estimated using the integrated SWAT-MODFLOW model. The SWAT-MODFLOW model provided estimates at both the HRU level and MODFLOW grid cells, and the total annual average groundwater recharge (in Mm^3) over the watershed was approximately 297.48. However, groundwater extraction directly affects the hydrological regime of the associated aquifer, minimising recharge and reducing base flow, and should not exceed recharge to supply aquatic environments. Exploitable groundwater is the annual amount of water abstracted from an aquifer, accounting for socioeconomic and environmental constraints (Sophocleous, 2000). The areal-averaged annual groundwater abstraction can be directly derived from the model area based on the specific yield and expected water level decline during dry periods to sustain the ecosystems. However, exploitable groundwater resources are obtained as the rate of annual average groundwater discharge, excluding the minimum desirable flow that can sustain the environment during low flow periods. Based on an assessment of stream-aquifer interactions, the total average annual groundwater discharge (in Mm^3) was calculated to be approximately 349.65.

Despite this, the quantity of groundwater that contributes to environmental streamflow and ecosystem services during low-flow settings is beneficial for the development of river changes. As proposed by Smakhtin (Smakhtin, 2001), the base flow can be considered equal to Q_{90} , and the monthly streamflow exceeds 90 % of the time. Thus, we computed the SGWI-simulated streamflow exceeding 90 % probability with the overall aim of quantifying the groundwater footprint as essential hydrologic information to properly address groundwater shed sustainability. We then determined the total average annual minimum desirable flow (in Mm^3) to be 13.72. During low-flow periods, when surface water is scarce for human needs, groundwater frequently supports people, riverine, and riparian habitats. Groundwater consumption during low-flow seasons affects aquatic life, especially in areas with scarce surface water and prevalent groundwater exploitation. Critical environmental streamflow limits caused by pumping and declining groundwater levels can result in severe economic and environmental consequences.

The analysis assumes that groundwater is withdrawn at sustainable rates, which are derived indirectly through the GFI rather than by directly estimating extraction rates. This approach captures the overall balance between natural recharge and groundwater use, providing insights into the sustainability of the system. While the model does not explicitly incorporate all aspects of groundwater use, such as detailed groundwater abstraction rates, demand-supply balances, and irrigation return flows, these are indirectly represented through recharge and baseflow dynamics. Owing to regional data limitations, we assumed that groundwater abstraction was reflected in the stress indicators derived from the model and that irrigation return flows were embedded within the surface water budget simulated by SWAT.

Evaluating watershed GF can improve the accuracy of sustainable groundwater management strategies. We found that evaluating the groundwater footprint is important for climate adaptation and informing aquifer system water balances for sustainable groundwater development. Overall, the GFI of the watershed was calculated to be approximately 1.23, which indicates moderate groundwater depletion and highlights the need for appropriate management strategies to avoid depletion. A GFI of less than one implies resilient groundwater use, whereas a footprint greater than one indicates non-resilient discharge, potentially impacting future groundwater and ecosystems, particularly in extreme environments. Hence, the area under consideration may threaten future regional economic growth and increase water resource projects if appropriate measures are not taken. Moderate groundwater depletion poses significant challenges to both ecological systems and human activities. This level of depletion suggests that extraction exceeds the natural recharge of the aquifer, indicating that stress on groundwater resources leads to potential long-term environmental consequences.

Quantifying GF using datasets derived from the calibrated coupled model can significantly reduce systematic gaps from data uncertainties. Although developing an integrated model that incorporates both surface and subsurface systems is data-intensive and time-consuming, our study provides reliable insights into the region. This indicates that the sustainability of the aquifer under consideration is better than that of most global aquifers. Our findings indicate that consumption surpasses the natural renewable water supply in watersheds. Unless managed sustainably, anthropogenic activities lead to a decline in groundwater storage during droughts, resulting in severe water stress in the area. Groundwater management requires holistic trade-offs (Mays, 2013) and minimises uncertainties in the groundwater footprint (Gleeson and Wada, 2013). A watershed-scale understanding of sustainable groundwater management is vital for adapting appropriate and robust sustainability assessments, resulting in unbiased mitigation strategies (Gohari et al., 2013) that prioritise watershed health without affecting ecosystem services.

3.5. Implications for aquifer management

The depletion of groundwater resources within ecosystems can adversely affect vegetation and streamflow, thereby affecting biodiversity and aquatic habitats. Decreasing groundwater can pressure vegetation, reduce plant coverage, alter habitat suitability, and increase vulnerability to environmental stressors. Reduced baseflow due to groundwater depletion can disturb aquatic habitats, affecting fish populations, biodiversity, and ecosystem health. Agriculture and other human activities reliant on groundwater face reduced water availability, which negatively affects productivity and livelihoods. Reduced water resources in agriculture lead to lower crop yields and changing cropping patterns, thereby affecting food security. In addition, it can exacerbate socioeconomic inequalities, particularly affecting the vulnerabilities of those who rely on groundwater. This poses a significant risk to drinking water availability, potentially leading to increased costs and challenges in ensuring water quality compliance. Hence, the results underscore the need for sustainable water resource management, balancing demand and supply, prioritising environmental health, and ensuring equitable access. Water scarcity affects residential supply, sanitation, and hygiene, thereby affecting marginalised communities. Groundwater depletion threatens ecosystems and biodiversity and alters vegetation and habitats in wetlands. Changes in streamflow affect aquatic ecosystems, fish populations, and ecological services. Groundwater fluctuations disturb wetlands, thereby affecting agricultural productivity negatively. Depletion exacerbates social inequities and intensifies disparities and conflicts regarding water availability.

This study underscores the broader impacts of groundwater management and advocates for strategies that consider hydrological and environmental factors. Policymakers should promote sustainable water management, ecosystem resilience, and community health by addressing groundwater depletion, degradation, and insecurity issues.

Therefore, assessing watershed system responses to groundwater consumption is essential for sustainable water management and long-term economic viability. Furthermore, we proposed and developed a novel groundwater footprint hydrological model application by exploring an ensemble approach to the integrated SWAT-MODFLOW model. Despite thorough calibration attempts, hydrological models inevitably contain errors owing to the intricate and nonlinear nature of the real systems. The robustness of the modelling framework enhances the accuracy of model predictions, enabling policymakers to make decisions based on solid evidence to protect groundwater resources, address urgent concerns and advance sustainable management. This study provides useful hydrological insights into groundwater practices, supporting the consideration of SGWI in water policy, agriculture, and ecosystem management in the region. We introduced a novel paradigm for understanding the water balance of watershed aquifers using the integrated SWAT-MODFLOW model to assess groundwater sustainability. Hence, the current novel ensemble approach improves spatiotemporal hydrologic assessments by enhancing data derivation frameworks for estimating groundwater footprints from the integrated SWAT-MODFLOW model.

Future research should examine the impacts of various water resource management scenarios on ecological systems. Furthermore, future studies should consider the effects of climate change when adapting water management practices to evolving social and economic conditions, potentially mitigating the risks associated with sustainable groundwater supply. An accurate depiction of hydrological processes is strongly dependent on data availability. Both the quantity and quality of input dataset pose significant limits to hydrological modelling, and the presence of limited data has the potential to introduce biases and uncertainties into model estimates. In this study, we focused solely on streamflow uncertainty analysis during model simulation because of the constraints posed by the absence of groundwater observation time series data. Overall, this study enhances our understanding of watershed hydrology by offering a novel ensemble approach that is useful for informing water resource management practices and guiding the sustainable utilisation of both surface water and groundwater resources.

4. Conclusions

In this study, we developed an integrated model for a watershed to analyse the long-term impacts of groundwater use under surface water-groundwater interactions (SGWI). Our study presents a novel ensemble approach for estimating groundwater footprints to support sustainable groundwater management. The SWAT-MODFLOW model was used to evaluate watershed hydrological system responses and examine the SGWI. Through calibration, we reduced the uncertainties related to aquifer features, thereby enabling more precise predictions. We estimated the ensemble values of the exploitable amounts of groundwater discharge, minimum desirable flow, and groundwater recharge to quantify the groundwater footprint of the watershed.

These outcomes suggest the need for aquifer management to secure a sustainable water supply for communities and ecosystems. Groundwater resource depletion can adversely affect vegetation and streamflow, thereby affecting biodiversity and aquatic habitats. Decreasing groundwater can pressure vegetation, reduce plant coverage, alter habitat suitability, and increase vulnerability to environmental stressors. A robust assessment of the SGWI is crucial for understanding and managing extreme hydrological responses. Groundwater footprint analysis, encompassing usage, renewal, and ecosystem needs, facilitates sustainability evaluation and informs decision-making processes. Our study significantly contributes to efficient groundwater resource management, paving the way for more comprehensive approaches in the future. The model highlights the role of river-aquifer exchange in maintaining environmental flow during dry periods, thus ensuring ecosystem health during droughts.

The applied applications of these results include supporting evidence-based policies to regulate groundwater abstraction, protect recharge zones, and develop drought contingency plans that sustain environmental flows. The model outputs provide a foundation for designing targeted interventions to safeguard groundwater-dependent ecosystems and agricultural livelihood. The ensemble modelling approach presented in this study offers a transferable framework for assessing groundwater sustainability in other data-scarce regions.

Future research should incorporate continuous groundwater-level monitoring and explore adaptive management scenarios that balance human demand with socio-ecological requirements. Overall, this study demonstrates how integrated ensemble modelling can inform evidence-based policies to protect groundwater resources and support resilient ecosystems. Therefore, assessing the spatio-temporal groundwater footprint of SGWI is essential for developing low-impact watershed management strategies that ensure long-term groundwater sustainability.

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CRediT authorship contribution statement

Sun Woo Chang: Software, Resources, Project administration, Funding acquisition. **Mingyu Kim:** Visualization, Resources, Data curation. **Il-Moon Chung:** Resources, Project administration, Funding acquisition. **Tarekegn Dejen Mengistu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization,

Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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